PYTHON IMPLEMENTATION IN BIOLOGY

Python implementation in biology means using the Python programming language to analyze biological data, simulate biological processes, study DNA & proteins, perform bioinformatics research, and apply Artificial Intelligence in life sciences.

Biology today generates huge datasets (DNA sequences, medical records, experimental results).
Python helps scientists process, analyze, visualize, and predict biological behavior quickly and accurately.

It is widely used in:

- Bioinformatics
- Genetics & genomics
- Medical research
- Drug discovery
- Ecology & environmental biology
- AI in healthcare

Why Python is Used in Biology

Python helps biologists to:

- Analyze DNA & protein sequences
- Study genetic mutations
- Model biological systems
- Analyze laboratory experiment data
- Predict disease patterns using AI
- Visualize biological processes

Popular libraries:

- **Biopython** – DNA & protein analysis
- **NumPy / Pandas** – biological data analysis
- **Matplotlib** – graphs & visualization
- **Scikit-learn** – AI & machine learning

Program 1 — DNA Length Finder

```python
dna = input("Enter DNA Sequence: ")
print("DNA Length:", len(dna))
```

Program 2 — DNA Base Counter

```python
dna = input("Enter DNA Sequence: ")

print("A:", dna.count("A"))
print("T:", dna.count("T"))
print("G:", dna.count("G"))
print("C:", dna.count("C"))
```

Program 3 — RNA Conversion

```python
dna = input("Enter DNA Sequence: ")

rna = dna.replace("T","U")
print("RNA Sequence:", rna)
```

Program 5 — Simple Bacterial Growth

```python
bacteria = int(input("Initial Bacteria Count: "))

for i in range(5):
    bacteria *= 2
    print("After Hour", i+1, ":", bacteria)
```

Program 6 — GC Content Analyzer

```python
dna = input("Enter DNA Sequence: ")

gc = dna.count("G") + dna.count("C")
gc_content = (gc/len(dna))*100

print("GC Content:", gc_content,"%")
```

Program 7 — Mutation Detection

```python
seq1 = input("Enter Original DNA: ")
seq2 = input("Enter Mutated DNA: ")

for i in range(len(seq1)):
    if seq1[i] != seq2[i]:
        print("Mutation at position", i)
```

Program 8 — Population Growth Model

```python
pop = int(input("Initial Population: "))
rate = float(input("Growth Rate: "))
years = int(input("Years: "))

for i in range(years):
    pop = pop*(1+rate)
    print("Year", i+1, ":", int(pop))
```

Program 9 — DNA Reverse Complement

```python
dna = input("Enter DNA: ")

comp = dna.replace("A","t").replace("T","a").replace("G","c").replace("C","g").upper()
print("Reverse Complement:", comp[::-1])
```

Program 10 — DNA Frequency Graph

```python
import matplotlib.pyplot as plt

dna = input("Enter DNA: ")

values = [dna.count("A"), dna.count("T"), dna.count("G"), dna.count("C")]
labels = ["A","T","G","C"]

plt.bar(labels, values)
plt.title("DNA Base Frequency")
plt.show()
```

Program 11 — DNA Transcription (DNA → mRNA)

```python
dna = input("Enter DNA Sequence: ")

mrna = dna.replace("A","U").replace("T","A").replace("G","C").replace("C","G")
print("mRNA Sequence:", mrna)
```

Program 12 — Amino Acid Counter

```python
protein = input("Enter Protein Sequence: ")

unique = set(protein)

for aa in unique:
    print(aa, ":", protein.count(aa))
```

Program 13 — Codon Splitter

```python
dna = input("Enter DNA Sequence: ")

for i in range(0, len(dna), 3):
    print("Codon:", dna[i:i+3])
```

Program 14 — BMI Calculator (Human Biology)

```python
weight = float(input("Enter Weight (kg): "))
height = float(input("Enter Height (m): "))

bmi = weight/(height**2)
print("BMI =", bmi)
```

Program 15 — DNA Palindrome Check

```python
dna = input("Enter DNA: ")

if dna == dna[::-1]:
    print("Palindrome Sequence")
else:
    print("Not Palindrome")
```

Program 16 — Species Population Comparison

```python
species1 = [10,15,20,25]
species2 = [12,18,22,30]

for i in range(len(species1)):
    print("Year",i+1,"Difference:",species2[i]-species1[i])
```

Program 17 — Logistic Population Growth

```python
pop = 50
r = 0.2
K = 500

for t in range(10):
    pop = pop + r*pop*(1-pop/K)
    print("Population:", int(pop))
```

Program 18 — DNA Similarity Percentage

```python
seq1 = input("DNA 1: ")
seq2 = input("DNA 2: ")

match = 0
for i in range(len(seq1)):
    if seq1[i] == seq2[i]:
        match += 1

print("Similarity:", (match/len(seq1))*100,"%")
```

Program 19 — Enzyme Reaction Rate

```python
substrate = float(input("Enter Substrate Concentration: "))
Vmax = 100
Km = 50

rate = (Vmax*substrate)/(Km+substrate)
print("Reaction Rate:", rate)
```

Program 20 — Ecology & Environmental Biology

```python
plants = 50
growth = 5

plants += growth
print("Plant population:", plants)
```

Project Title

AI-Based Disease Prediction System using Python

Aim

To predict the possibility of a disease based on biological/medical parameters using machine learning.

Concept

The system analyzes biological parameters like:

- Age
- Blood Pressure
- Sugar Level
- Heart Rate
- BMI

and predicts:

Disease risk (Yes / No)

Used in:

- Medical diagnostics
- Healthcare AI
- Predictive biology

Algorithm

1. Create biological dataset
2. Select features & label
3. Train ML model
4. Predict disease risk
5. Display result

```python
from sklearn.tree import DecisionTreeClassifier

# Sample biological dataset
# [Age, BP, Sugar, HeartRate]
X = [
    [45,140,200,90],
    [25,120,85,70],
    [50,150,220,95],
    [30,110,90,72],
    [60,160,240,100]
]

# Disease labels
y = [1,0,1,0,1]   # 1 = Disease, 0 = No Disease
```

```python
# Disease labels
y = [1,0,1,0,1]   # 1 = Disease, 0 = No Disease

# Model
model = DecisionTreeClassifier()
model.fit(X,y)

print("Enter Patient Data")

age = int(input("Age: "))
bp = int(input("Blood Pressure: "))
sugar = int(input("Sugar Level: "))
hr = int(input("Heart Rate: "))

prediction = model.predict([[age,bp,sugar,hr]])

if prediction[0] == 1:
    print("⚠️ Disease Risk Detected")
else:
    print("✅ No Disease Risk")
```